Jia-Huei (Dylan) Ju

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EDUCATION

Johns Hopkins University — Baltimore, MD, USA
Ph.D. Student in Computer Science — Sep. 2026 – Present
(advisors: Dr. Andrew Yates and Prof. Benjamin Van Durme)

University of Amsterdam — Amsterdam, Netherlands
Ph.D. Student at Informatics Institute (transferred to JHU) — Mar. 2024 – Aug. 2026
(advisors: Dr. Andrew Yates, Prof. Suzan Verberne, and Prof. Maarten de Rijke)

National Chengchi University — Taipei, Taiwan
M.Sc. in Management Information Systems — Sep. 2019 – Sep. 2021
(advisors: Prof. Hao-Chun Chuang and Prof. Yen-Chun Chou)

RESEARCH EXPERIENCE

HLTCOE, Johns Hopkins University — Baltimore, MD, USA
Visiting Researcher | Supervisors: Dr. Eugene Yang and Dr. Andrew Yates — Jun. 2025 – Aug. 2025

- Designed a document re-ranking method for optimizing information coverage.
- Developed nugget-based retrieval methods for retrieval-augmented generation.

IRLab, University of Amsterdam — Amsterdam, Netherlands
Ph.D. Student | Supervisors: Dr. Andrew Yates and Prof. Maarten de Rijke — Mar. 2024 – Aug. 2026

- Developed coverage-aware dense retrieval with coverage-based learning via augmented training data.
- Designed a retrieval evaluation method for assessing the performance of long-form retrieval-augmented generation.
- Integrated learned sparse retrieval into multimodal and multilingual information retrieval tasks.

CFDA Lab, Research Center for IT Innovation, Academia Sinica — Taipei, Taiwan
Research Assistant | Supervisor: Dr. Chuan-Ju Wang — Dec. 2019 – Jan. 2024

- Designed a multi-view learning framework for an ad-hoc passage re-ranking model.
- Developed a data augmentation method for training a conversational passage re-ranking model.
- Constructed relevance-aware question generators for improving out-of-domain text ranking.
- Defined a new signal highlighting task with a retrieval pipeline for financial analysis.

PUBLICATIONS

Peer-Reviewed

[1] **Search for Coverage: Learning Coverage-Aware Retrieval with Augmented Sub-Question Answerability**
Jia-Huei Ju, Eugene Yang, Trevor Adriaanse, Suzan Verberne, and Andrew Yates — *SIGIR 2026*

[2] **LANCER: LLM Reranking for Nugget Coverage**
Jia-Huei Ju, François G. Landry, Eugene Yang, Suzan Verberne, and Andrew Yates — *ECIR 2026*

[3] **MILCO: Learned Sparse Retrieval Across Languages via a Multilingual Connector**
Thong Nguyen, Yibin Lei, **Jia-Huei Ju**, Eugene Yang, and Andrew Yates — *ICLR 2026*

[4] **SERVAL: Surprisingly Effective Zero-shot Visual Document Retrieval Powered by Large Vision and Language Models**
Thong Nguyen, Yibin Lei, **Jia-Huei Ju**, and Andrew Yates — *EMNLP 2025*

[5] **Controlled Retrieval-Augmented Context Evaluation for Long-Form RAG**
Jia-Huei Ju, Suzan Verberne, Maarten de Rijke, and Andrew Yates — *EMNLP 2025*

[6] **Reproducibility, Replicability, and Insights into Visual Document Retrieval with Late Interaction**
Jingfen Qiao, **Jia-Huei Ju**, Xinyu Ma, Evangelos Kanoulas, and Andrew Yates — *SIGIR 2025*

[7] **Relevance-aware Diverse Query Generation for Out-of-domain Text Ranking** *(NVIDIA collaboration)*
Jia-Huei Ju, Chao-Han Yang, Szu-Wei Fu, Ming-Feng Tsai, and Chuan-Ju Wang — *RepL4NLP @ ACL 2024*

[8] **Improving Conversational Passage Re-ranking with View Ensemble**
Jia-Huei Ju, Sheng-Chieh Lin, Ming-Feng Tsai, and Chuan-Ju Wang — *SIGIR 2023*

[9] **A Compare-and-contrast Multistage Pipeline for Uncovering Financial Signals in Financial Reports**
Jia-Huei Ju, Yu-Shiang Huang, Cheng-Wei Lin, Che Lin, and Chuan-Ju Wang *ACL 2023*

[10] **Text-to-Text Multi-view Learning for Passage Re-ranking**
Jia-Huei Ju, Jheng-Hong Yang, and Chuan-Ju Wang *SIGIR 2021*

Preprints

[1] **Relevance Ranking as a Path to Automated Relevance Judgment**
Jia-Huei Ju, Eugene Yang, Suzan Verberne, and Andrew Yates *Preprint, 2026*

[2] **DICE: Dual-Granularity Index for Claim-Enhanced Retrieval**
Jia-Huei Ju, Eugene Yang, Suzan Verberne, and Andrew Yates *Preprint, 2026*

[3] **Learned Nugget Assignments**
Jia-Huei Ju, Lukas Gienapp, Martin Potthast, Eugene Yang, Andrew Yates, and Harrisen Scells *Preprint, 2026*

PROFESSIONAL SERVICE

Reviewing

Program Committee	SIGIR'25–26; ECIR'26
Reviewer	ARR (Oct & Dec'24, Feb & May'25); EMNLP'22–23; ICLR'22; ICML'23; NeurIPS'23; SIGIR'23; WWW'22–24

Teaching

Guest Lecturer	Information Retrieval 1 (2025)
Teaching Assistant	Information Retrieval 1, Information Retrieval 2 (2025); Recommender Systems (2026)

Student Supervision

Thesis	Emmanouil Georgios Lionis (MSc, 2024); Tobias Peeters (BSc, 2025)
Projects	*The Powerless Noise: How Experimental Settings Shape the Reported Power of Noise* (SIGIR'26 Reproducibility) *On the Reproducibility of Learned Sparse Retrieval Adaptations for Long Documents* (ECIR'25 Reproducibility)